

Sharp Consensus-Time Asymptotics for a Degenerate Monotone Aggregation Rule

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Abstract

Consider asynchronous, with-replacement aggregation on N binary agents with rule $f(x_1, x_2, x_3) = x_1 \vee (x_2 \wedge x_3)$. If T_N is the absorption time and $u_N(s) = \mathbb{E}[T_N \mid S_0 = s]$, we determine the sharp $N^{3/2}$ asymptotics of the worst initial state. An exact Green kernel and a uniform boundary-layer analysis give a limiting profile U . Its unique maximizer x_* is characterized by equality of two convergent integrals, and

$$N^{-3/2} \max_{1 \leq s \leq N} u_N(s) \longrightarrow C_* = 0.94149933731 \dots$$

Every finite- N maximizing state, including either member of a possible tie, satisfies $s/\sqrt{N} \rightarrow x_* = 0.86314241094 \dots$. The decimal values are only numerical orientations; the theorem gives exact integral formulas.

1 Statement and context

At each discrete time, choose an updating agent uniformly from $[N]$ and, independently, choose three input agents uniformly with replacement. The updating bit becomes

$$f(x_1, x_2, x_3) = x_1 \vee (x_2 \wedge x_3),$$

and all other bits remain fixed. Thus one unit of time is one asynchronous update. Write S_t for the number of ones,

$$T_N = \inf\{t \geq 0 : S_t \in \{0, N\}\}, \quad u_N(s) = \mathbb{E}[T_N \mid S_0 = s].$$

Mossel's general analysis of monotone aggregation dynamics gives order $\Theta(N^{3/2})$ for the worst expected time in this residual-minterm case and supplies a general finite Green-function framework [1, Theorem 1(iii), Proposition 13, Sections 2.6–2.7]. The result below identifies the constant, the full scaled profile in the $\sqrt{N}$ boundary layer, and all subsequential limits of maximizers. This is a comparison with the closest inspected result, not a claim of literature priority.

Put

$$\Phi(x) = \int_0^x e^{-t^2/2} dt, \quad S = \Phi(\infty) = \sqrt{\frac{\pi}{2}},$$

and, for $x > 0$, define

$$A(x) = \int_0^x \frac{\Phi(y)e^{y^2/2}}{y} dy,$$

$$B(x) = \int_x^\infty \frac{(S - \Phi(y))e^{y^2/2}}{y} dy.$$

The integrand of A is continuously extended to have value 1 at zero.

Theorem 1. *There is a unique $x_* > 0$ such that*

$$A(x_*) = B(x_*).$$

For every compact $K \subset [0, \infty)$,

$$\sup_{x \in K} \left| \frac{u_N(\lfloor x\sqrt{N} \rfloor)}{N^{3/2}} - U(x) \right| \rightarrow 0,$$

where $U(0) = 0$ and, for $x > 0$,

$$U(x) = \frac{S - \Phi(x)}{S} A(x) + \frac{\Phi(x)}{S} B(x).$$

Consequently,

$$\lim_{N \rightarrow \infty} N^{-3/2} \max_{1 \leq s < N} u_N(s) = C_* = A(x_*) = B(x_*).$$

Moreover, every choice $s_N \in \arg \max_{1 \leq s < N} u_N(s)$ satisfies $s_N/\sqrt{N} \rightarrow x_$. Hence the full limit set in the extended half-line of scaled maximizing states is $\{x_*\}$; in particular, it does not contain $+\infty$.*

The exact characterization may alternatively be written with standard special functions:

$$A(x) = S \int_0^x \frac{e^{y^2/2} \operatorname{erf}(y/\sqrt{2})}{y} dy, \quad B(x) = S \int_x^\infty \frac{\operatorname{erfcx}(y/\sqrt{2})}{y} dy.$$

2 Exact finite chain

Conditional on $S_t = s$, set $p = s/N$. The three sampled bits are independent Bernoulli variables with parameter p , even when sampled indices coincide, because the indices themselves are independent. The mean of the rule is $g(p) = p + p^2 - p^3$. Thus the nonzero jump probabilities of (S_t) are

$$\beta_{N,s} = p(1-p)(1+p-p^2), \tag{1}$$

$$\delta_{N,s} = p(1-p)(1-p^2), \tag{2}$$

and holding has the remaining probability. Hence $u_N(0) = u_N(N) = 0$ and

$$\beta_{N,s}(u_N(s+1) - u_N(s)) + \delta_{N,s}(u_N(s-1) - u_N(s)) = -1. \tag{3}$$

For $1 \leq j < N$, let

$$r_{N,j} = \frac{\delta_{N,j}}{\beta_{N,j}} = \frac{1 - (j/N)^2}{1 + j/N - (j/N)^2}, \quad q_{N,1} = 1, \quad q_{N,k} = \prod_{\ell=1}^{k-1} r_{N,\ell},$$

and set

$$H_{N,s} = \sum_{k=1}^s q_{N,k}, \quad H_N = H_{N,N}, \quad Q_{N,s} = H_N - H_{N,s}.$$

Lemma 2 (Green kernel and increments). *For $1 \leq s, j < N$, the expected number of visits to j before absorption, counting time zero, is*

$$G_N(s, j) = \frac{H_{N, \min(s, j)} Q_{N, \max(s, j)}}{H_N \delta_{N, j} q_{N, j}}. \tag{4}$$

Writing $a_{N,j} = (\delta_{N,j} q_{N,j})^{-1}$, one has

$$u_N(s) = \frac{Q_{N,s}}{H_N} \sum_{j=1}^s H_{N,j} a_{N,j} + \frac{H_{N,s}}{H_N} \sum_{j=s+1}^{N-1} Q_{N,j} a_{N,j}. \quad (5)$$

If

$$A_N(s) = \sum_{j=1}^s H_{N,j} a_{N,j}, \quad B_N(s) = \sum_{j=s}^{N-1} Q_{N,j} a_{N,j},$$

then, for $1 \leq s \leq N$,

$$u_N(s) - u_N(s-1) = \frac{q_{N,s}}{H_N} \{B_N(s) - A_N(s-1)\}. \quad (6)$$

The expression in braces is strictly decreasing in s .

Proof. As a function of s , the right-hand side of (4) vanishes at 0 and N and is harmonic away from j . Since $\beta_{N,j} q_{N,j+1} = \delta_{N,j} q_{N,j}$, applying the generator at its pole gives

$$-\frac{\beta_{N,j} H_{N,j} q_{N,j+1} + \delta_{N,j} Q_{N,j} q_{N,j}}{H_N \delta_{N,j} q_{N,j}} = -1.$$

Uniqueness of the finite Dirichlet problem proves (4); summing the columns proves (5).

Let $d_{N,s} = u_N(s) - u_N(s-1)$. Equation (3) yields

$$\frac{d_{N,s+1}}{q_{N,s+1}} = \frac{d_{N,s}}{q_{N,s}} - a_{N,s}.$$

Together with $\sum_{s=1}^N d_{N,s} = 0$, summing and exchanging the two finite sums gives (6). Its brace, denoted $F_N(s)$, obeys the exact identity

$$F_N(s+1) - F_N(s) = -H_N a_{N,s} < 0.$$

Thus u_N is unimodal, with at most two adjacent tied maximizers. $\square$

3 Uniform boundary-layer limit

Write $R(p) = (1 - p^2)/(1 + p - p^2)$. The following estimates contain the uniformity needed later to pass to the maximum.

Lemma 3. *For each fixed $L < \infty$,*

$$\sup_{1 \leq k \leq L\sqrt{N}} \left| \log q_{N,k} + \frac{k(k-1)}{2N} \right| = O_L(N^{-1/2}), \quad (7)$$

whereas globally

$$q_{N,k} \leq \exp \left\{ -\frac{2k(k-1)}{5N} \right\}. \quad (8)$$

Consequently, locally uniformly for $x \geq 0$,

$$\frac{H_{N, \lfloor x\sqrt{N} \rfloor}}{\sqrt{N}} \longrightarrow \Phi(x), \quad \frac{H_N}{\sqrt{N}} \longrightarrow S. \quad (9)$$

Furthermore, locally uniformly on $(0, \infty)$,

$$\frac{A_N(\lfloor x\sqrt{N} \rfloor)}{N^{3/2}} \longrightarrow A(x), \quad (10)$$

$$\frac{B_N(\lfloor x\sqrt{N} \rfloor)}{N^{3/2}} \longrightarrow B(x). \quad (11)$$

The first convergence extends locally uniformly to $x = 0$.

Proof. Uniformly for bounded $p\sqrt{N}$, Taylor expansion gives $\log R(p) = -p + O(p^2)$; summation proves (7). Also

$$1 - R(p) = \frac{p}{1 + p - p^2} \geq \frac{4p}{5},$$

so $R(p) \leq e^{-4p/5}$ and summation gives (8). The local estimate, Gaussian domination from the global estimate, and Riemann sums yield (9).

If $j/\sqrt{N} \rightarrow y > 0$, then (7) and (2) give

$$\frac{a_{N,j}}{\sqrt{N}} \longrightarrow \frac{e^{y^2/2}}{y}. \quad (12)$$

Equations (9) and (12) therefore give (10) by Riemann summation on intervals away from zero. On $j \leq L\sqrt{N}$, one has $q_{N,j} \geq c_L$, $\delta_{N,j} \geq c_L j/N$, and $H_{N,j} \leq j$. The normalized summands are thus uniformly bounded, so an initial interval of scaled length ε contributes $O_L(\varepsilon)$. This proves the asserted extension to zero.

It remains to justify the infinite upper limit in (11). Direct differentiation gives

$$R'(p) = -\frac{1 + p^2}{(1 + p - p^2)^2} < 0.$$

Monotonicity therefore provides the geometric estimate

$$\frac{Q_{N,j}}{q_{N,j}} \leq \frac{r_{N,j}}{1 - r_{N,j}} = \frac{1 - (j/N)^2}{j/N}. \quad (13)$$

Together with (2), this implies

$$\frac{Q_{N,j}}{\delta_{N,j} q_{N,j}} \leq \frac{N}{j} + \frac{N^2}{j^2} + \frac{N}{N - j}. \quad (14)$$

Hence, for $s \geq 2$,

$$B_N(s) \leq \frac{2N^2}{s - 1} + 2N(1 + \log N), \quad (15)$$

and in particular

$$\lim_{L \rightarrow \infty} \limsup_{N \rightarrow \infty} \frac{B_N(\lfloor L\sqrt{N} \rfloor)}{N^{3/2}} = 0.$$

On a fixed window $x\sqrt{N} \leq j \leq L\sqrt{N}$, the same Riemann-sum argument as before applies; the last display removes the truncation. This proves (11). The limiting integral is finite because Mills' inequality bounds its integrand by y^{-2} . $\square$

Proof of Theorem 1. Substituting (9)–(11) into (5) proves the claimed profile convergence on compact subsets of $(0, \infty)$. To include zero, let $s \leq \sqrt{N}$. The bounds used above control the first sum in (5); splitting the second sum at $\lfloor \sqrt{N} \rfloor$ and applying (15) to the remainder gives, with an absolute C ,

$$\frac{u_N(s)}{N^{3/2}} \leq C \left[\frac{s}{\sqrt{N}} + \frac{s}{\sqrt{N}} \left(1 + \log \frac{\sqrt{N}}{s+1} \right) \right]. \quad (16)$$

The right-hand side tends uniformly to zero first for $s \leq \varepsilon \sqrt{N}$ and then as $\varepsilon \downarrow 0$. The same is true of $U(x)$ by its integral formula. This proves compact-uniform convergence on $[0, \infty)$.

Now A is strictly increasing from 0 to ∞ , whereas B is strictly decreasing from ∞ to 0; the behavior of B at infinity again follows from Mills' inequality. Their crossing x_* therefore exists and is unique. Differentiation of U , with cancellation of the two endpoint terms, gives

$$U'(x) = \frac{e^{-x^2/2}}{S} \{B(x) - A(x)\}. \quad (17)$$

Thus U has its unique maximum at x_* and $U(x_*) = A(x_*) = B(x_*)$.

Finally, (10), (11), and (6) show locally uniformly for $x > 0$ that

$$N^{-3/2} F_N(\lfloor x\sqrt{N} \rfloor) \longrightarrow B(x) - A(x),$$

where one-index shifts have size $o(N^{3/2})$ on such compact intervals. Given $0 < \varepsilon < x_*$, this limit is positive at $x_* - \varepsilon$ and negative at $x_* + \varepsilon$. Because F_N is strictly decreasing, every maximizer of u_N is trapped between the corresponding two scaled indices for all sufficiently large N , including both indices in a possible adjacent tie. Hence every maximizing sequence satisfies $s_N/\sqrt{N} \rightarrow x_*$. Compact-uniform profile convergence then gives convergence of the maximum to $U(x_*)$. The same trapping excludes $+\infty$ from the extended-half-line limit set. $\square$

4 Numerical orientation and methodological note

Quadrature of the exact integrals gives

$$x_* = 0.8631424109396626\dots, \quad C_* = 0.941499337310331\dots$$

These values are not inputs to the proof. The decisive ingredients are the uniform tail estimate (15) and the exact increment identity (6). In a different probability problem, Zeng likewise combines a uniform coefficient estimate with exact verification of a remaining finite range [2]. This is a proof-architecture comparison only and supports no claim about consensus dynamics.

Computational and AI assistance disclosure. Generative AI assisted with proof exploration, algebraic cross-checking, and drafting. Deterministic scripts compared finite tridiagonal Poisson solutions with the Green formula and evaluated the displayed decimals. These were checks only; the proof is analytic and self-contained. The author reviewed the argument and takes responsibility for the manuscript.

References

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